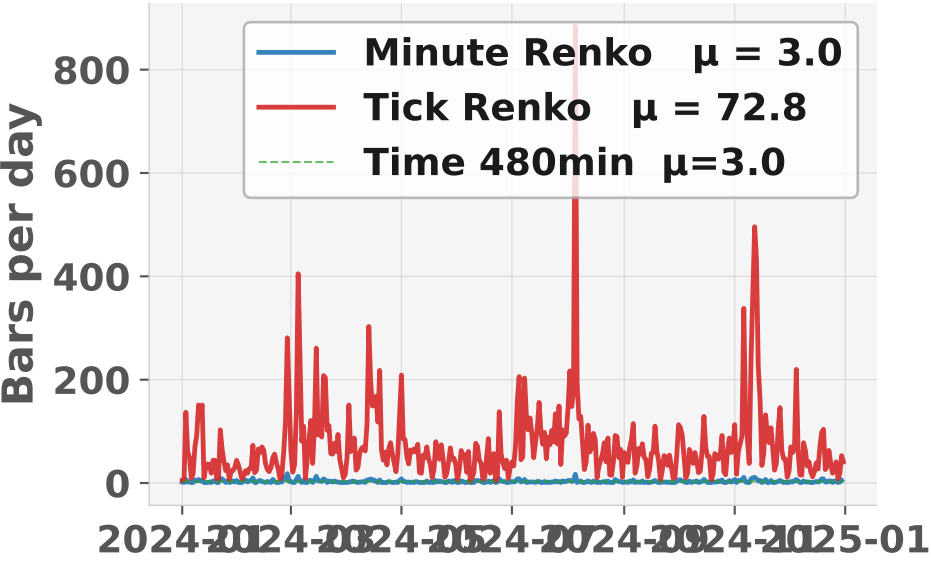
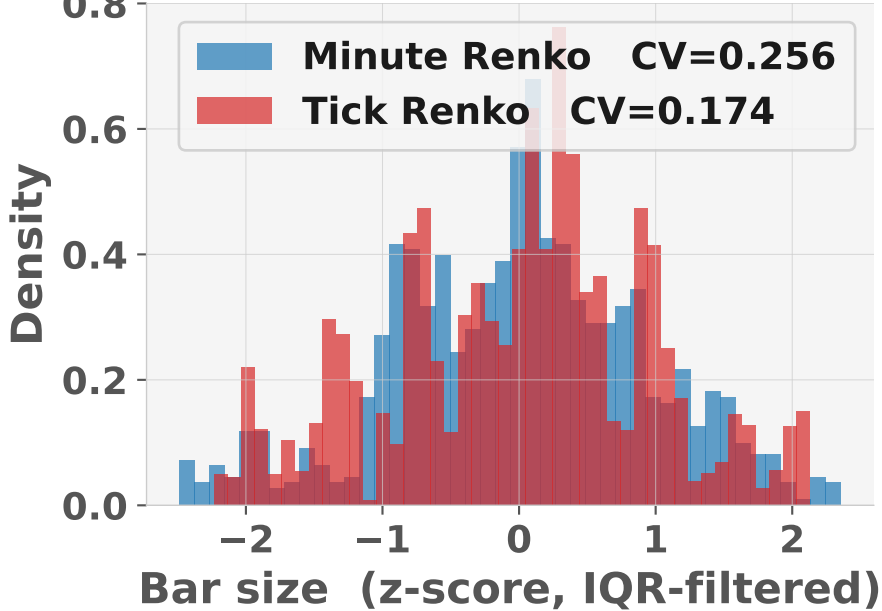
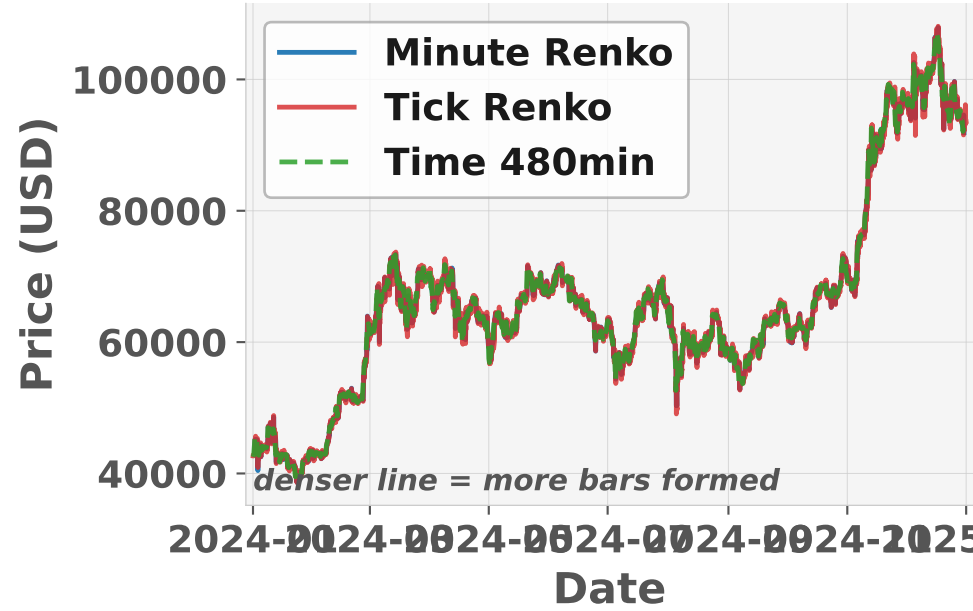
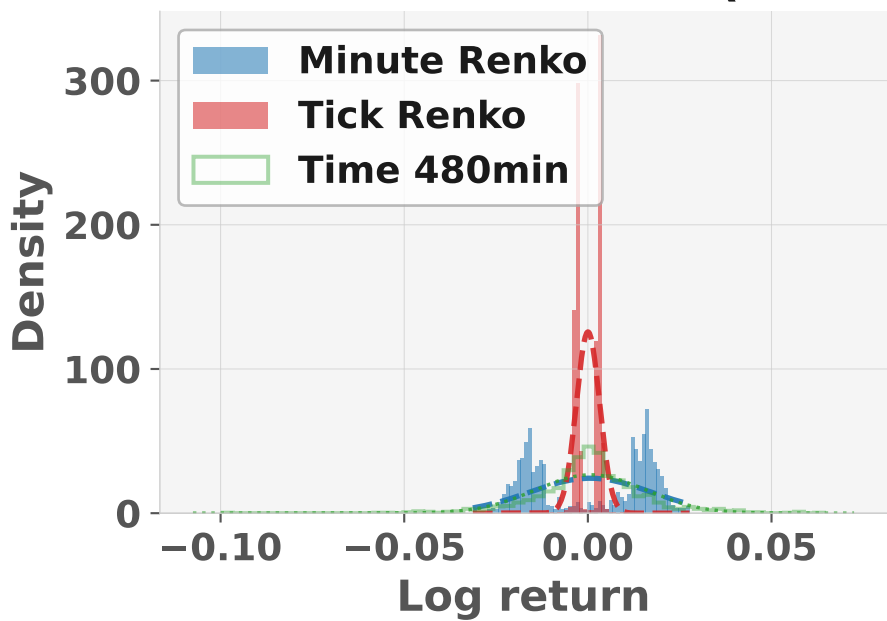


Information Bar Comparison: Minute Renko vs Tick Renko vs Time 480min (time bar baseline)

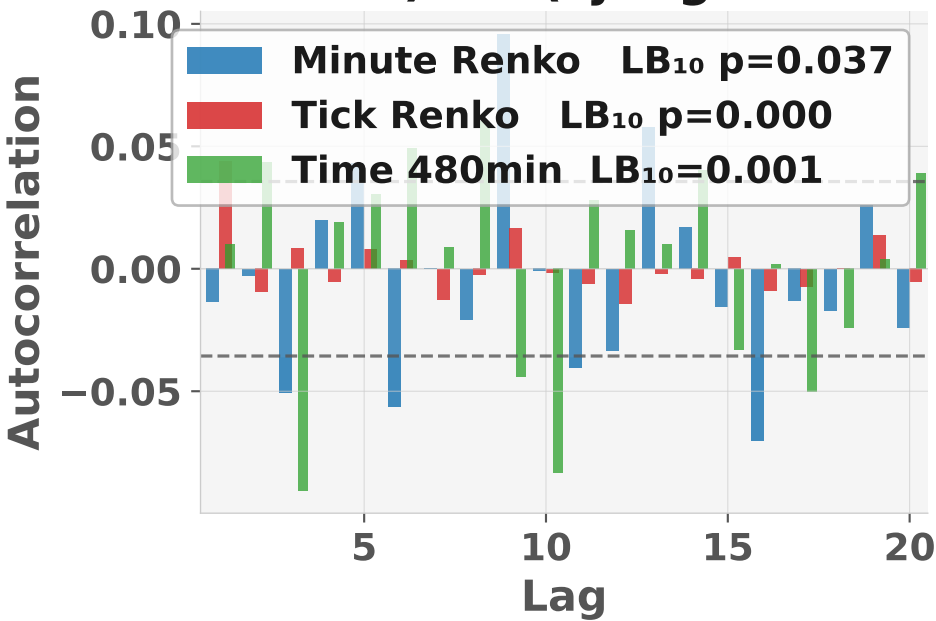
Close Price — A: 3.0 bars/day · B: 72.8 bars/day · CV lower = more uniform · Bare Formation Frequency Tick Renko · Variation with market activity = good



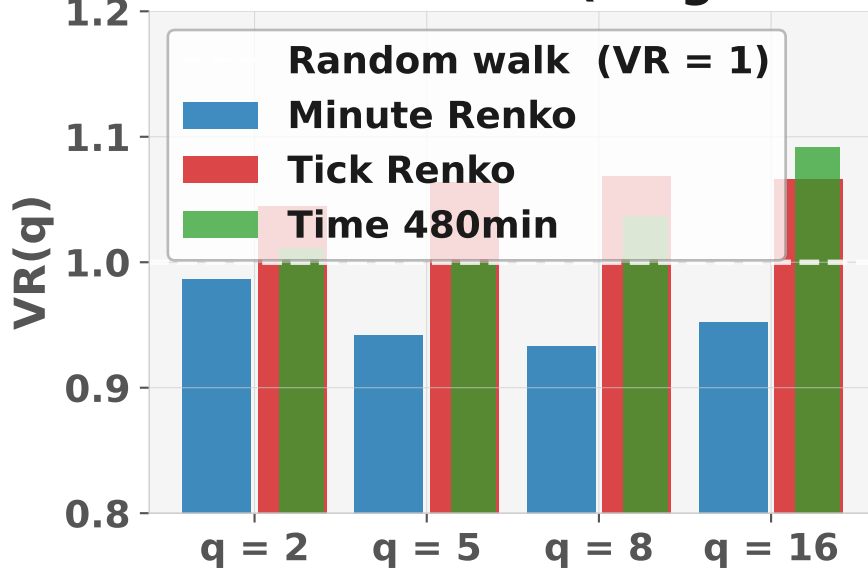
Return Distribution (dashed = fitted normal)



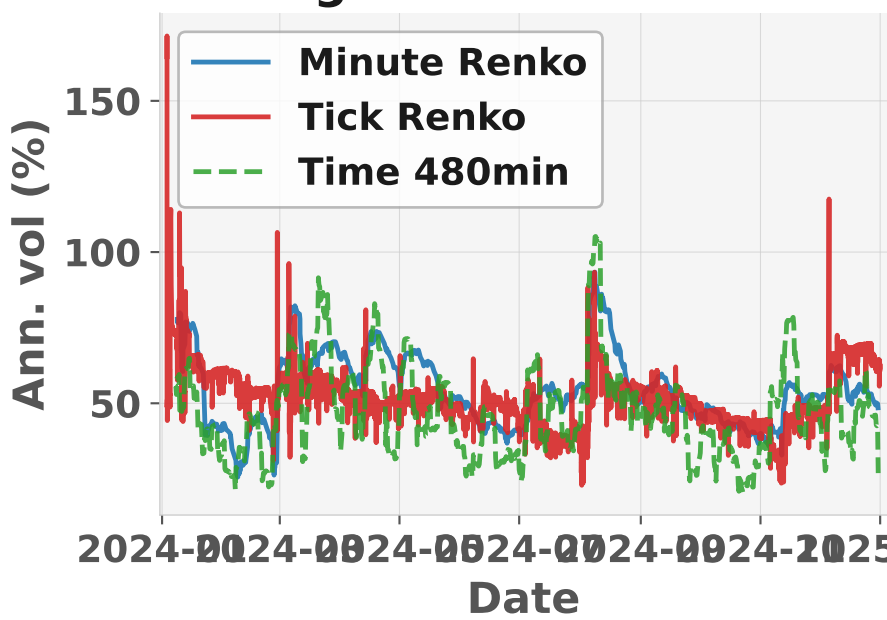
Return ACF (Ljung-Box test)



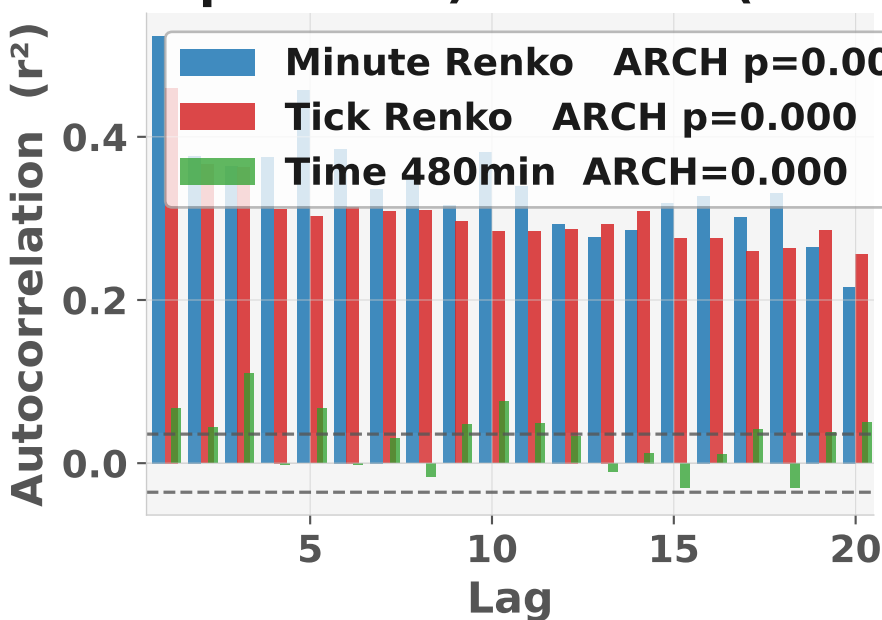
VR(5) = 1 - |AC lag-1| · A = 1 bar cluster ratio (vs. random walk) · B = 1 bar cluster ratio (vs. random walk) · C = 1 bar cluster ratio (vs. random walk)



Rolling Annualised Volatility (window = 20 days)



Return ACF (aligned with Minute Renko, leads on 3 of 8 criteria)



Less clustering = better · leads: ≈ tie

Criterion	Minute Renko	Tick Renko	Time
Bar size CV	0.256	0.174	n/a
Entropy	4.348	2.718	4.088
Kurtosis	1.443	1.391	3.553
AC lag-1	0.013	0.044	0.010
LB p (10)	0.037	0.000	0.001
JB stat	99.440	2168.950	581.110
VR(5)-1	0.058	0.063	0.000
Timeout %	10.500	0.100	n/a

Green = leads on this criterion · Yellow = tie
Leading on more criteria ≠ better trading performance